

Technology Stack

Date	6 th July 2026
Team ID	SWTID-2026-4078
Project Name	Credit Card Approval Prediction
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Flask	Provides a responsive and userfriendly interface for entering applicant details and displaying prediction results
2	Backend / Server-Side	Python, Flask, Scikit-learn	Handles business logic, processes user inputs, loads the trained machine learning model, and generates real-time credit card approval predictions.
3	Database / Data Storage	SQLite / CSV Dataset	Stores applicant information and historical data efficiently with lightweight database management.
4	Cloud / Hosting / Deployment	IBM Watson Machine Learning, Flask	Deploys the trained machine learning model on the cloud and enables secure real-time prediction through a web application.
5	Version Control & CI/CD	Git, GitHub	Maintains version control, enables team collaboration, tracks project changes, and simplifies project maintenance.
6	Third-Party APIs / Other Tools	Pandas, NumPy, Matplotlib, Joblib	Supports data preprocessing, numerical computations, data visualization, and machine learning model serialization for deployment.

